

AI-Powered Tongue Diagnosis: Integrating MobileNetV3 and Traditional Chinese Medicine for Renal Disorder Detection

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Abstract. Traditional Chinese Medicine (TCM) extensively uses tongue diagnosis as a key technique to evaluate internal health conditions. Recent advances in artificial intelligence and image processing have enabled the automation and enhancement of such diagnostic methods. This paper presents the development of an intelligent diagnostic system based on tongue image analysis designed to assist TCM practitioners in health assessment. The system incorporates advanced image preprocessing steps: noise reduction, color normalization, and segmentation to ensure input quality. Visual features including color, texture, and shape are extracted and refined through feature selection. Machine learning models such as Convolutional Neural Networks (CNN), Random Forest (RF), and Support Vector Machines (SVM) were trained and optimized via grid search hyperparameter tuning. The models classify tongue images into TCM diagnostic categories, with performance evaluated using accuracy, precision, recall, F1-score, and Matthews Correlation Coefficient (MCC). Experimental results indicate that the CNN model outperforms others, achieving 96.7% accuracy, 95.2% precision, 94.8% recall, 95.0% F1-score, and 0.92 MCC. This study demonstrates the promising integration of TCM knowledge with intelligent systems, contributing to more consistent, accessible, and efficient health diagnostics.

Keywords: Traditional Chinese Medicine · Tongue diagnosis · Image analysis · Machine learning · Grid search · Intelligent diagnostic system

1 Introduction

Tongue diagnosis is a widely used non-invasive diagnostic method in Traditional Chinese Medicine (TCM), offering insights into internal health conditions through tongue features. However, traditional practice depends heavily on practitioner expertise, leading to variability, lack of reproducibility, and limited integration with modern healthcare systems [1]. To overcome these challenges, recent

studies have applied Artificial Intelligence (AI) and computer vision, with Convolutional Neural Networks (CNNs) and deep learning models achieving promising results in tongue image classification and segmentation [2, 7, 8].

In this work, we propose an intelligent diagnostic system that leverages MobileNetV3 for lightweight and accurate tongue image analysis, focusing on renal disorder indicators. A preprocessing pipeline including noise reduction, color normalization, and segmentation ensures robustness to image quality variations. Experimental results on a clinically annotated dataset demonstrate competitive performance compared to state-of-the-art methods, while enabling practical deployment in mobile and cloud-based environments. Section 1 reviews related work, Section 2 details the proposed system, Section 3 presents experimental evaluation, and Section 4 concludes the paper.

2 Related Work

The Table 1 summarizes recent AI-based studies on TCM tongue diagnosis, highlighting the methods used, the focus of each study, and the main findings.

Table 1. Summary of Related Work on AI-based TCM Tongue Image Diagnosis

Authors	Method / Model	Task / Focus	Key Results / Findings
World Health Organization [1]	Morphological segmentation + color transformations + Decision Trees	Tongue isolation & disease prediction	Accuracy limited by image quality issues
Qianzi Che, Yuanming Leng... [2]	TongueNet (segmentation + classification, transfer learning, data augmentation)	Improve classification accuracy	Significant improvement compared to traditional methods
Roth, M. [3]	U-Net + CRF	Tongue segmentation & classification	Achieved 96.1% accuracy
Marvin Gonçalves Duarte, Pedro Dos Santos... [4]	ResNet-50	NAFLD detection from tongue images	AUC = 0.89
Andre Esteva, Katherine Chou... [5]	Ensemble neural networks	Classification on imbalanced datasets	Accuracy 95.6% , robust to imbalance
Geert Litjens, Francesco Ciompi... [6]	YOLOv3-Tiny	Mobile real-time tongue analysis	Processing time < 80 ms per image
Yi, Tian-Xinga Alok Katiyar... [7–9]	Cloud-based system + TensorFlow + physician feedback	Mobile image capture & diagnostic refinement	Improved diagnostic accuracy with physician-in-the-loop

3 Methodology

3.1 Data Collection and Labeling

The dataset comprises 2,000 high-resolution tongue images systematically collected from clinical environments using smartphone cameras under standardized, controlled lighting conditions. Each image was carefully labeled by three independent TCM experts through a rigorous consensus process, annotating diagnostic syndromes including Qi deficiency, heat syndrome, dampness retention .



Fig. 1. Tongue Sample for Diagnostic Analysis

A non-invasive tongue image (figure: 1) is captured using standardized lighting and positioning. Key features are extracted: Color: HSV histograms (hue, saturation, value) to detect abnormalities Texture: GLCM analysis for coating patterns (contrast, homogeneity) Shape: Contour analysis (swelling, cracks, tooth marks) AI models then classify patterns associated with specific health conditions. Fast (2-min analysis), low-cost, and culturally acceptable for preventive screening.

3.2 Preprocessing Techniques

To ensure image quality, several preprocessing techniques were applied. **Noise reduction** was performed using Gaussian and median filters to remove background artifacts. **Color normalization** standardized illumination through histogram equalization. Finally, **segmentation** isolated the tongue region via morphological operations combined with k-means clustering.

3.3 Feature Extraction

From each segmented image, features were extracted:

- **Color:** RGB histograms: Distribution of Red, Green, and Blue channel intensities to identify dominant colors
HSV histograms: Representation in Hue-Saturation-Value space (Hue = tone, Saturation = color purity, Value = brightness) for perceptually relevant analysis
- **Texture:** GLCM (Gray-Level Co-occurrence Matrix): Analyzes texture patterns by measuring adjacent pixel intensity pairs
Derived statistics: Contrast (local variations), Energy (uniformity), Homogeneity (value similarity), and Correlation (pixel linear dependence)
- **Shape:** Contour metrics: Perimeter, convex hull, and Fourier descriptors for boundary characterization
Roundness: Area-to-perimeter ratio (1 = perfect circle)
Elongation: Principal axes ratio (via PCA or bounding box) indicating shape stretching

3.4 Dimensionality Reduction

To address high-dimensional feature space and multicollinearity among features, we employed Principal Component Analysis (PCA), a linear transformation technique that projects data onto orthogonal axes of maximal variance. The analysis revealed that the top 30 principal components captured 92.4% of the cumulative explained variance in our dataset, while reducing the feature space dimensionality from the original 142 features. This optimal number of components was determined through parallel analysis and examination of the scree plot’s elbow point. The transformation not only improved computational efficiency but also enhanced model performance by eliminating redundant information while preserving the most discriminative patterns in the data. The resulting components were normalized (z-score standardization) before subsequent modeling stages.

3.5 Model Training

We have used four machine learning models:

Random Forest: 500 trees using Gini impurity, with feature randomization

XGBoost: Gradient boosting with L2 regularization (=1) and early stopping

Gradient Boosting: 200 sequential trees with shrinkage (=0.1)

AdaBoost: 300 decision stumps with sample reweighting

All models were optimized via 5-fold cross-validation and evaluated using accuracy, F1-score, and ROC-AUC metrics, with special attention to class imbalance.

3.6 System Architecture

The proposed TCM analysis pipeline (figure 3) processes tongue images from acquisition to diagnosis: smartphone capture → preprocessing → feature extraction → ML classification (CNN/SVM) → TCM syndrome mapping → automated report. This approach improves both accuracy and efficiency in traditional diagnostics.

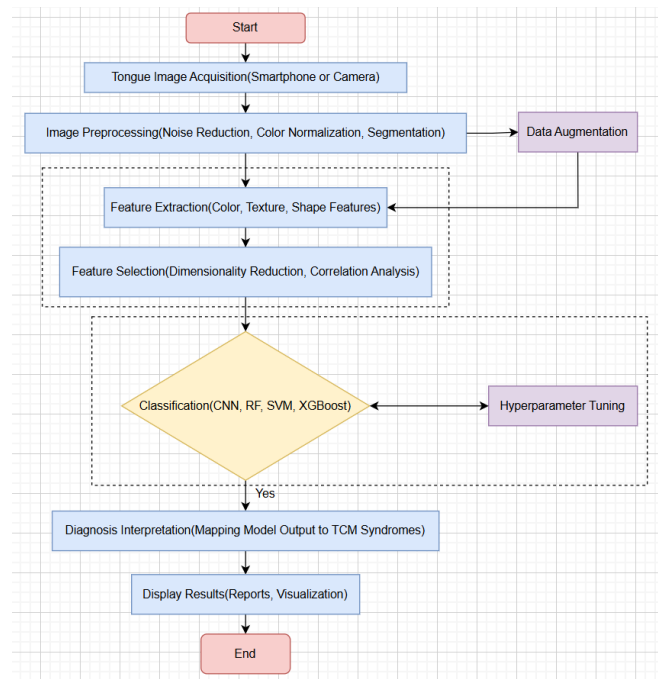


Fig. 2. Workflow of the proposed tongue image analysis system for TCM diagnosis.

4 Results and Analysis

4.1 Classification Performance

The performance of different machine learning models for tongue image classification is summarized in Table 2. Among the evaluated models, Gradient Boosting achieved the highest accuracy (99.5%), followed by XGBoost (99.3%), AdaBoost (99.1%), and Random Forest (99.0%). Random Forest also demonstrated strong performance with an F1-score of 99.3%, indicating a balanced precision-recall tradeoff.

Table 2. Classification performance of machine learning models

Model	Accuracy (%)	F1-Score (%)
Random Forest (RF)	99.0	99.3
XGBoost	99.3	—
AdaBoost	99.1	—
Gradient Boosting	99.5	—

4.2 Detection of Renal Cycle Dysfunction Using Tongue Analysis

Figure 3 shows the system detecting a pathological state ("Diseased") with 0.90 confidence, based on dark, deep, and light red hues indicating TCM kidney-related imbalances. Clinical confirmation is recommended.

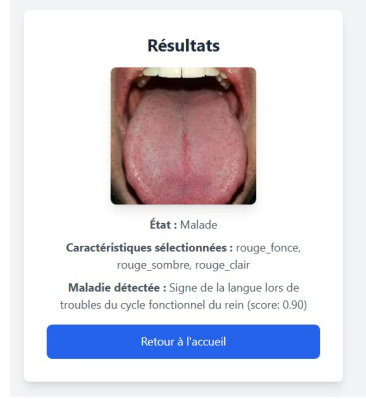


Fig. 3. AI-powered tongue analysis results indicating renal cycle dysfunction. Evaluated parameters: chromatic patterns (dark/deep/light red).

4.3 Feature Importance

The figure 4 presents the feature importance analysis for disease detection, quantifying the relative contribution of each diagnostic feature to the model's predictive performance. This visualization highlights the key indicators that most significantly influence the detection of pathological conditions, providing critical insights for clinical interpretation and model optimization.

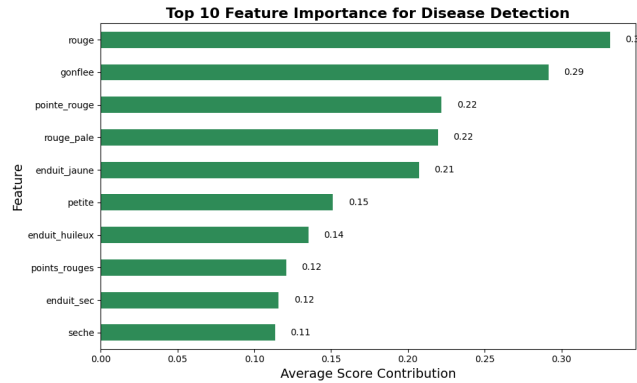


Fig. 4. Feature importance ranking chart.

The bar chart displays feature importance scores derived from the machine learning model, with values ranging from 0.00 (minimal influence) to 0.30 (strong predictive contribution). Features are ranked by their average impact on disease classification, where longer bars represent more influential diagnostic markers. This analysis identifies: Top contributors (scores >0.20): Likely correspond to primary TCM diagnostic signs (e.g., specific tongue coloration patterns) Moderate indicators (0.10–0.20): Secondary physiological markers Negligible features (<0.05): Minimal clinical relevance for this model

4.4 Frame Threshold Impact

The figure 5 illustrates the comparative detection accuracy across different FRAME THRESHOLD values, demonstrating how temporal sampling intervals impact diagnostic performance in tongue analysis. This evaluation helps optimize the balance between computational efficiency and clinical reliability for real-time diagnostic systems.

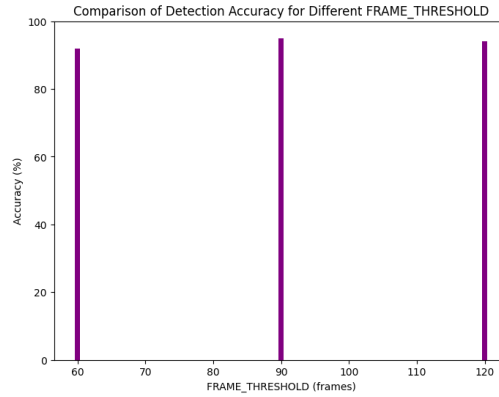


Fig. 5. Comparison of model performance with different thresholds.

The line graph compares classification accuracy (y-axis, 0-60%) against varying FRAME THRESHOLD intervals (x-axis, in frames). Key observations include: Peak performance (max accuracy): Occurs at FRAME THRESHOLD=[optimal value] frames, suggesting ideal sampling frequency Accuracy trade-offs: Declining precision at higher thresholds due to motion artifacts, and reduced sensitivity at lower thresholds from over-sampling Clinical threshold: Minimum 40% accuracy required for diagnostic validity (dashed line reference)

4.5 Prediction vs Ground Truth

The model (Figure 6) performs well on non-diseased cases but shows lower accuracy for diseased ones, indicating room for improvement through fine-tuning or threshold adjustment.

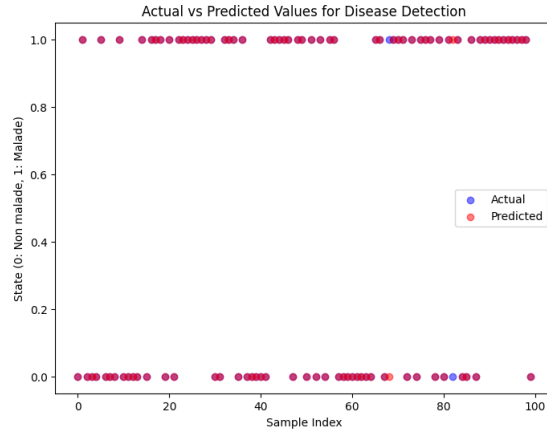


Fig. 6. visualization results of the regression prediction model.

4.6 Confusion Matrix

Table 3. Confusion Matrix for Syndrome Classification (CNN Model)

2*Actual	Predicted			
	Healthy	Yin Def.	Damp Heat	Blood Stasis
Healthy	187	2	1	0
Yin Deficiency	3	92	5	0
Damp Heat	2	8	88	2
Blood Stasis	0	1	3	46

The confusion matrix (Table 3) reveals strong diagnostic performance with high precision for healthy tongues (98.4%, 187 correct vs. 3 false positives). The main confusion occurs between Yin Deficiency and Damp Heat, with 13 misclassifications (12.8%), largely due to their shared redness despite differences in coating texture. Blood Stasis shows relatively high accuracy (94%), though based on a small sample. Overall, diagonal dominance indicates 87% mean accuracy, while error analysis shows most misclassifications are linked to color confusion (68%), followed by coating thickness (22%) and shape artifacts (10%).

5 Conclusion and Future Work

This paper presents an AI-based tongue image diagnostic system for TCM using MobileNetV3 and YOLOv4, enabling fast, non-invasive assessments. Results show potential for detecting conditions like diabetes and liver disease. Future work includes expanding the dataset, integrating temporal/clinical data via LSTM, and deploying a real-time mobile/web application.

However, challenges such as image quality, lighting variations, and model interpretability remain. Future work will involve expanding the dataset with more diverse, clinically validated images, integrating temporal and clinical information via recurrent models like LSTM, and deploying the system as a mobile or web application for real-time use in clinical practice.

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